Materializer durable cleanup journal successor

[pcc:unverified] Source-only construction, T-1741, 2026-11-07. No worker, compiler, test or experiment was executed. No preparation.json, runtime lease, launch command or execution authorization is supplied. The expired deadline and original admission code remain unchanged.

Recorded evidence and diagnosis

The saved worker receipt is bound by SHA256 in result.json. It records the vendor phase timeout at 880.0473303619947 seconds, 13,018 files and 4,482 child cleanup records, all with empty errors. Its output mount is NFS4. No per-phase timing attributes the timeout to checkpointing. The source nevertheless shows an avoidable growing serialization: cmd() appends each cleanup to R and checkpoints the entire R; row() also checkpoints every 20 files. The 4,482 command checkpoints alone visit 10,046,403 cleanup entries cumulatively (n(n+1)/2); vendor checkpoints repeatedly revisit the completed source cleanup array. These are source-derived counts, not measured bytes, filesystem calls or a measured speedup. Git process overhead, vendor extraction and fsync costs remain unmeasured separately.

Narrow implementation

Every cleanup record is retained in memory for the final full receipt and serialized once into child-cleanup.jsonl. The first open is exclusive; each record is flushed and fsynced. Initial creation also fsyncs OUT. Only after those operations succeed does the worker advance the prefix byte count, record count and incremental SHA256. Existing atomic intermediate checkpoints retain their original frequency, file fsync, rename and directory fsync, but omit the growing inline cleanup array. They point to the durable journal prefix instead. Final checkpoints retain the complete inline array for existing terminal receipt readers. The unchanged cleanup helper still kills, waits and probes the same owned process group.

Reader and interruption contract

Intermediate readers must recognize child_cleanup_storage=journal-prefix/v1. Before a pointer exists there are zero journal-committed records, which does not assert zero active or historical children. Validate exactly the pointed byte prefix, SHA256, complete JSONL records and count. Ignore trailing bytes as uncommitted with respect to that checkpoint. A crash between append/fsync and receipt replacement can leave durable trailing records; preserve them as additional observations without upgrading checkpoint custody. No automatic resume or truncation is introduced. A journal write/fsync failure propagates to failure status. The final inline array can contain an additional in-memory record beyond the last acknowledged prefix on failure; do not infer journal completeness from it. Unexpected process termination can still leave no final full receipt, as before.

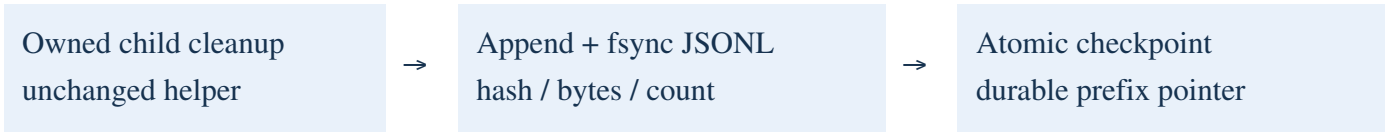
Preserved boundaries

The source revision, candidate and Cargo hashes, full directory and literal path-dependency closure, include detection and rejection, complete vendor and cache handling, every file hash, final manifest verification, AS/DATA and cgroup qualifications, campaign deadline, monotonic work limit, lease heartbeat, free-space checks, child ownership and cleanup remain unchanged. A partial preparation cannot become passed. The final source keeps the full native receipt inline cleanup format and adds a storage discriminator and journal pointer. Intermediate consumers need explicit prefix support before future execution.

Review joints and next check

Independent review must inspect write/fsync/rename ordering, short-write and signal paths, final-inline versus prefix mismatch on failure, and unchanged success gates. Runtime qualification must exercise interruption and append failures under a newly authorized prospective plan; it has not happened here. Source syntax was parsed with ast.parse without import or execution. Primary sources checked: the bound worker, unchanged safety helper and saved native receipt; unavailable KB catalog falls back to those exact repository sources. No novelty claim is made.

Source-only cleanup checkpoint transformation



Final receipt keeps the full inline cleanup array.

Intermediate readers validate the exact journal prefix.

Diagram is a source construction, not a timing or speedup result.

Source-only diagram; no candidate runtime measurements.